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Project Title:	Cyclonic Two-Phase Flow Separator Experiment in Microgravity		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Comments:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kadambi, Jaikrishnan Ph.D. (Case Western Reserve University)		
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Task Description:

NOTE that this is a Flight Experiment follow-on to "Investigation into the Quantification and Control of Steady and Unsteady Flow Phenomena in Passive Gas-Liquid Cyclonic Separators" grant NNX09AI31G with same principal investigator.
Multiphase flow separation is a critical technological challenge for the human exploration of space and planetary bodies. While droplet and bubble segregation is a spontaneously occurring phenomenon in most terrestrial situations, the absence of the gravity-induced separatory action in a microgravity environment often results in situations where two disparate phases have no distinct inclination to separate from one another. This can present many problems with respect to spacecraft and space station operations as proper control and sequestration of the individual phases is often essential to the successful functionality of many system-critical components.

Passive cyclonic separators are a promising technology that potentially could fulfill the separation needs of many of these systems. While considerable research has been performed on this technology, the limitations of terrestrial testing, aircraft testing, and modeling necessitates long-term experimentation to raise these devices to a sufficient technology readiness level (TRL) to permit use within spacecraft systems. It is a lack of basic engineering and scientific knowledge that currently prevents these systems from being employed. At a basic engineering level, there is a distinct lack of data regarding both the steady and unsteady operational performance of vortex separators.

Issues such as where the designs can be expected to work and how to go about optimizing designs are unresolved. Accurate prediction of these phenomena using computational fluid dynamics (CFD) is currently not tenable due to the dearth of available data and incomplete modeling techniques. Beyond engineering, there is also a lack of knowledge regarding the fundamental multiphase fluid physics phenomena that occur within passive cyclonic separators. Issues such as gas core adjustment to injection flow rates, bubble dynamics, vortex flow interaction with capillary forces at the gas core, etc. are all unresolved. Inasmuch, this research effort seeks to resolve these issues. The main objectives are to:

1. Quantitatively assess and delimit the range of operability of a prototype passive cyclonic separator including the transient and steady-state instability behavior of the device with respect to both device and system-level conduct.

2. Elucidate and quantify the multiphase fluids physics phenomena governing the performance of microgravity two-phase separators.

3. Validate and verify the multiphase CFD modeling approaches that have been developed to simulate the complicated two-phase transport processes associated with phase separation in microgravity.

4. Validate and verify a hydraulic system code for the simulation of two-phase flow in microgravity.

To achieve these objectives, combined experimental, numerical, and phenomenological modeling analyses are being performed to quantitatively describe the performance and response of systems containing separators. Specifically, the work entails the creation of multiphase flow system that is able to provide metered quantities of gas and liquid into a test separator. The separator and the associated flow loop had been operated under a variety of specially tailored conditions, including specific combinations of parameters to examine flow features of interest including the steady-state performance margins of the separator, the transient response of the separator, vapor core instability and misting, system startup/shutdown, and system level performance and instabilities. The products of this research have included the flight experiment (including two-phase flow loop and passive cyclonic separator), validated and verified CFD modeling techniques, a microgravity hydraulic system code, and archival data capturing the important fluid physics associated with microgravity separator operations.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

From a microgravity/scientific standpoint, the experimentation, modeling, and analytical work of this research will serve to:
1. Provide elucidation and understanding of the above-mentioned two-phase flow phenomena associated with the operation of separators in multiphase systems

2. Provide details of the columnar stability of the gas core

3. Provide details of the time constants associated with the response of passive cyclonic separators

4. Validate and verify a coupled CFD and hydraulic system code for simulations of two-phase flow in microgravity

The resulting insight into phase separation, distribution, and control will help to afford NASA designers the latitude to take greater advantage of the benefits offered by the use of these multiphase systems in spacecraft applications. The experimental results will also facilitate further new and original design modifications to improve the performance of microgravity phase separation devices. Validation and verification of the numerical results are expected to permit the use of CFD as a reliable design tool to further assist in the creation and analysis of new separator designs, while the results from the system code are expected to assist in the creation and analysis of general multiphase systems in microgravity.

Overall, the research will help contribute to the fundamental scientific understanding of two-phase fluid behavior, both in microgravity and 1-g environments.

The passive cyclonic separator has long been a prominent technology for gas-liquid separation in microgravity due to its high efficiency and long-term operability. However, the complex flow physics within the separator presents great challenges in designing the device and investigating its performance. The current work utilize numerical simulations and analytical modeling approaches to construct a system of control volume equations that can quantitatively describe the key features of the separator, as well as defining the parameters for design purposes.
Numerical Modeling: Computational simulations in 3D and 2D domains have been performed using OpenFOAM. The Volume of Fluid model is used to capture the large scale gas-liquid interface, and the turbulent flow field is simulated with Reynolds Stress Transport Model. Both 3D and 2D simulations compare well with the experimental results. A parametric study is then conducted using 2D simulations to correlate the dimensionless parameters affecting the performance of the separator. The CFD technique generates enough amount of data to study to aid the derivation of the control-volume equations.

Task Progress:	Analytical Modeling: The analytical modeling was developed based on control-volume approximations. This approach helped clarify the fluid physics by transforming the complicated physical phenomena into simple control-volume equations representing the mass and angular momentum conservations as well as pressure balance at the interface. With the aid of CFD results, the control-volume equations are completed with the capability of predicting separator performance in regards to various geometric parameters and flow conditions. The predictions given by the control-volume equations are reasonably accurate compared to the ground-based experimental data. The swirl number is also correlated with other controlled parameters, and its value is found to be above unity, which indicates that the separator should be able to maintain its performance in microgravity. The control-volume equations were further improved to include gas-liquid two-phase injection situations by modifying the skin friction coefficient and the backpressure. Homogeneous two-phase flow is assumed at the injection for simplification. Comparisons are made between control-volume predictions and experimental results and showed reasonably good agreement. In addition, the surface tension effect was also investigated as it plays an important role at low flow rates in the microgravity environment. The injection Weber number has been included in the pressure balance equation. By decreasing the injection Weber number, the gas core eventually collapses, which leads to a critical Weber number for a stable gas core. The critical Weber number is determined analytically, and CFD results have confirmed the trend going towards the critical point as Weber number is decreased.
Bibliography Type:	Description: (Last Updated: 03/06/2024)
Abstracts for Journals and Proceedings	Li Y, Kamotani Y. "Study of passive cyclonic two-phase separator for space applications." Presented at the 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Li Y, Kamotani Y. "Control-volume study of flow field in a two-phase cyclonic separator in microgravity." Theor. Comput. Fluid Dyn. 2023 Feb;37(1):105-27. https://doi.org/10.1007/s00162-022-00635-w , Feb-2023
Dissertations and Theses	Li Y. (Yeyuan Li) "Numerical and Analytical Study of Passive Cyclonic Separator Performance in Microgravity." Dissertation, Case Western Reserve University, Cleveland, OH, January 2021. , Jan-2021